Functional Requirements Specification (FRS)

Project: Proof of Delivery (POD) Application — Version 1.2

Application Name:	Proof of Delivery (POD) App	**Version:**	1.2
Developed By:	Lokesh Pawalia & Sarthak Srivastava	**Date:**	August 31, 2026
Frontend Tech:	Flutter SDK (Dart / Android)	**Backend Tech:**	FastAPI (Python 3.11) + MongoDB
Cloud Hosting:	Railway Production Cloud	**Media CDN:**	Cloudinary Digital Asset Management

1. Executive Summary & Objectives

The **Proof of Delivery (POD) App (Version 1.2)** is an enterprise-grade mobile and cloud verification ecosystem designed to track, manage, and cryptographically audit the physical distribution of government and organizational subsidies to beneficiaries (farmers).

The system completely prevents distribution fraud, identity impersonation, and phantom beneficiary leakage by enforcing an unalterable multi-stage handover protocol: biometric face verification, real-time OTP matching, multi-angle photographic and videographic proofs, GPS timestamping, and immediate digital invoice reconciliation.

2. User Hierarchy & Roles

- **Delivery Partner (Field Agent):** Operative responsible for physical transit, client credential validation, biometric facial capture, and digital proof submission.
- **Beneficiary (Farmer):** Authorized recipient of subsidized equipment or goods, verified through registered credentials and OTP verification.
- **Central Administrator:** Administrative stakeholder monitoring live distribution audit logs, partner allocation, and Cloudinary media inspection.

3. Functional Requirements by Module

Module 1: Partner Authentication & Identity Onboarding

FR-1.1 Mobile Number & OTP Verification: The system authenticates field operatives through a 10-digit mobile number and numeric One-Time Password verification (POST /auth/send-otp and POST /auth/verify-otp).

FR-1.2 Mandatory 8-Step Profile Completion Sequence: Newly registered partners must complete an 8-field structured profile enforcing strict real-time data integrity before accessing delivery orders:

1. **Full Name:** Restricted strictly to alphabetic characters and spaces via software keyboard filter (FilteringTextInputFormatter.allow(RegExp(r'[a-zA-Z\s]'))). Numbers and special characters are physically prevented.

Length: 2 to 50 characters.

2. Email Address: Enforces RFC-compliant email formatting (^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$). Whitespace characters are blocked.

3. PIN Code: Exactly 6 numeric digits. Triggers an automated asynchronous background lookup to the Indian Postal API (https://api.postalpincode.in/pincode/{PINCODE}) upon entry of the 6th digit.

4. City & State: Automatically populated from the resolved PIN code data. Configured as read-only to eliminate manual entry mistakes, with an edit-toggle for edge-case overrides.

5. Address: Physical street/residential address with a mandatory minimum 5-character threshold.

6. Vehicle Type: Selection dropdown: Two Wheeler, Three Wheeler, Four Wheeler, or Other (defaults to Two Wheeler).

7. Vehicle Number: Automatic uppercase transformation (UpperCaseTextFormatter). Validated against Indian Motor Vehicle standards (e.g. DL 01 AB 1234, UP 16 CP 6755, or Bharat Series).

8. Aadhaar Number: Exactly 12 numeric digits. Follows UIDAI specifications prohibiting prefixes starting with 0 or 1 (^[2-9][0-9]{11}$).

FR-1.3 Profile Photo Cloud Sync: Delivery partners must upload a valid facial profile photo, transmitted as multipart/form-data to PATCH /auth/profile and hosted securely on Cloudinary.

Module 2: Beneficiary & Delivery Management

FR-2.1 Scoped Beneficiary Allocation: Operatives only view records assigned to their unique partner identifier (delivery_partner_id), categorized dynamically as All, Pending, or Delivered.

FR-2.2 On-Demand Beneficiary Creation: Field operatives can onboard new farmers with detailed metadata (Name, Village, District, PIN, GPS coordinates, itemized goods, and reference photograph).

FR-2.3 Direct Calling & Route Navigation: Beneficiary cards integrate direct system telephony (tel:) and geospatial navigation (geo:) to facilitate field operations.

Module 3: Multi-Stage Proof of Delivery (POD) Protocol

To eliminate proxy collections, delivery completion requires passing five consecutive verification gates:

- **Gate 1: Beneficiary OTP Handover:** Recipient provides the unique 4-digit SMS handover code generated for their subsidy parcel.
- **Gate 2: On-Device Facial Biometrics:** Google ML Kit executes local landmark detection and facial matching against the beneficiary's registered reference image without sending unencrypted biometric data over the wire.
- **Gate 3: Multi-Angle Photographic Proof:** Operative captures 1 to 5 high-resolution photos documenting the recipient receiving the subsidy goods in their physical context.
- **Gate 4: Video Handover Audit:** Live video recording (up to 30 seconds) documenting physical item transfer and audible confirmation.
- **Gate 5: Geotagging & Geofencing Stamp:** Device GPS coordinates are stamped with exact date/time and embedded into the immutable audit record.

Module 4: Automated Invoicing & PDF Inspection

FR-4.1 Automated Digital Invoice: Immediately upon delivery clearance, the system compiles a tamper-evident PDF invoice containing tracking IDs, items disbursed, partner details, recipient signature/photo, and GPS timestamps.

FR-4.2 Cloud Upload & In-App Native Inspection: Invoices are synchronized to Cloudinary (POD-App/invoices) and displayed natively within the application via Syncfusion PDF Viewer with pan, zoom, and local download capabilities.

4. Backend API Specification Matrix

Method	Endpoint	Description	Payload	Auth
POST	/auth/check-phone	Check if phone exists	JSON	Public
POST	/auth/send-otp	Dispatch SMS OTP	JSON	Public
POST	/auth/verify-otp	Verify partner OTP	JSON	Public
POST	/auth/login-or-register	Generate JWT session	JSON	Public
PATCH	/auth/profile	Update profile & image	Multipart / JSON	Bearer
GET	/auth/me	Current partner record	None	Bearer
GET	/farmers	List assigned deliveries	Query Params	Bearer
POST	/farmers	Register new farmer	Multipart (Form+Img)	Bearer
POST	/farmers/{id}/upload_proof	Submit POD proofs & invoice	Multipart Data	Bearer

5. Non-Functional Requirements (NFR)

• **Performance:** Sub-second REST latency under standard 4G/LTE; streaming multipart upload resilience with 120s timeout for remote field conditions.

• **Security & Compliance:** Passlib Bcrypt salt-hashing, standard HS256 JWT bearer authentication, and Aadhaar UIDAI non-prefix validation.

• **High Availability:** Railway high-concurrency Uvicorn engine backed by MongoDB Atlas cluster failover and Cloudinary redundant media storage.

6. Document Approvals & Sign-Off

Role	Name	Status	Sign-Off Date
Lead Developer	Lokesh Pawalia	**Approved**	August 31, 2026
Lead Developer	Sarthak Srivastava	**Approved**	August 31, 2026
Release Version	POD App Version 1.2	**Production Ready**	August 31, 2026